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APPLIED DATA SCIENCE PROJECT · CHECKPOINT 1

Content Safety Classification for Generative AI Systems

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11 November 2026

Problem

Generative AI needs a safety check between model and user, but today’s checks fall short.

01

Harmful content passes

Generative models can answer harmful requests or produce harmful output that reaches the user.

02

Harmless content is blocked

Existing guards often flag harmless requests as unsafe (false positives, over refusal), making the system less useful.

03

One policy does not fit all

Each dataset and organisation defines harm differently, so a ready made guard may not fit a specific policy.

Value Proposition

For teams that deploy generative AI systems and need to **stop harmful output without blocking legitimate content**, our project builds and benchmarks an **open, measurable content safety classifier** that makes **every safety decision transparent and traceable**.

How the Classifier Works

It checks both the prompt and the model's response before anything reaches the user.

INPUT

PROMPT
"How do I pick a lock?"

RESPONSE
"Sure. First you need..."

CLASSIFIER

Llama Guard 3
with a LoRA adapter

- Harm taxonomy is written into the prompt
- LoRA adapts it to the target policy

OUTPUT

SAFE
Passes through unchanged

UNSAFE
+ violated categories
e.g. S2 Non-Violent Crimes
Withheld or regenerated

Baseline for comparison: DeBERTa encoder fine tuned per category

Goal and Objectives

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PROJECT GOAL

Build and benchmark an open content safety classifier that stops harmful output without blocking legitimate content.

OBJECTIVES

01 **Align the taxonomies** of public safety datasets (BeaverTails, WildGuardMix)

02 **Test existing safety models** on our data before any training (zero shot)

03 **Fine tune a guard model** (Llama Guard 3) with LoRA on the target policy

04 **Benchmark** accuracy, false positives and over refusal against a DeBERTa baseline

Sustainable Development Goals



SDG 9

Industry, Innovation and Infrastructure

An open, measurable safety layer for the responsible adoption of generative AI.

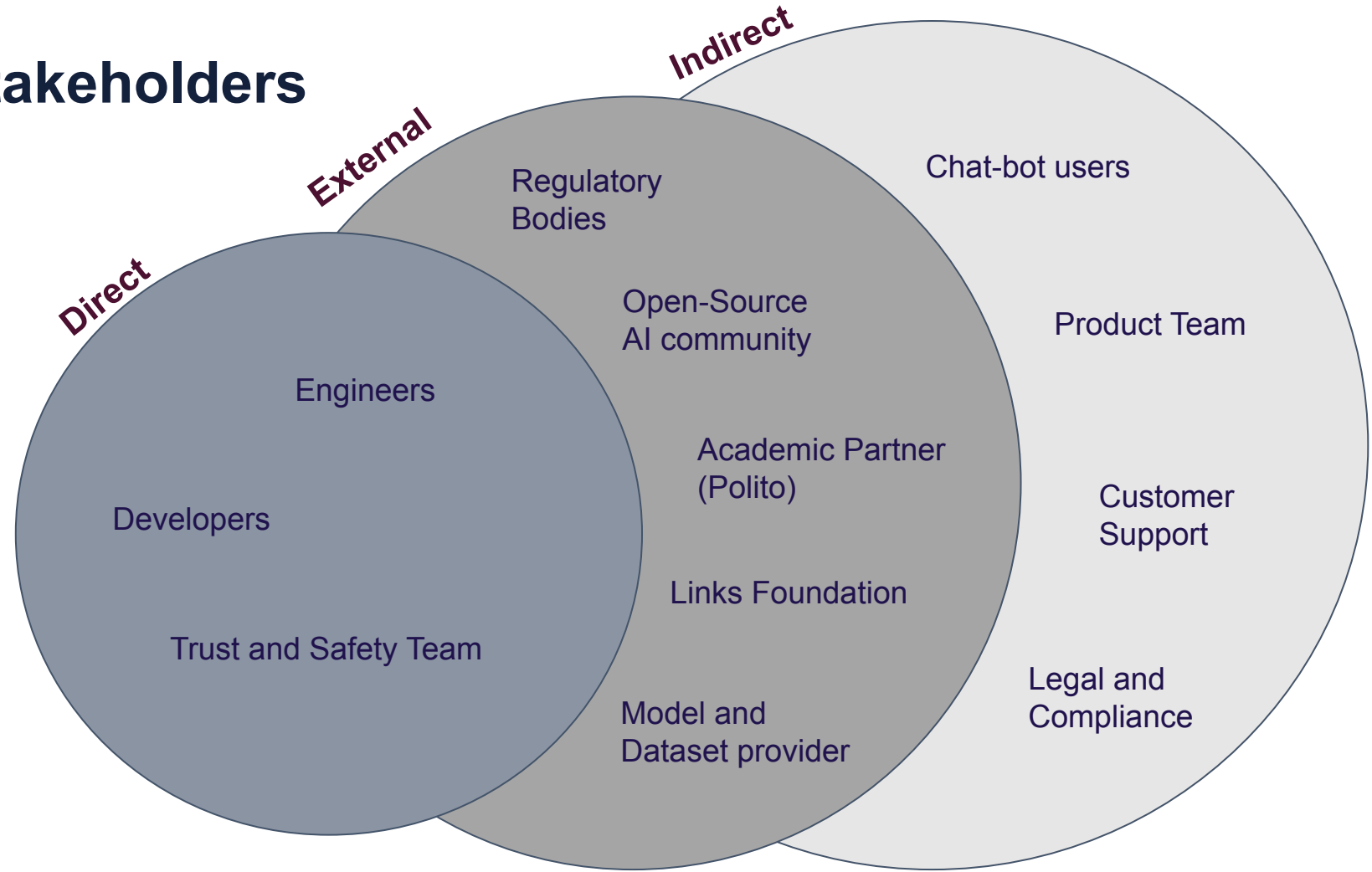


SDG 16

Peace, Justice and Strong Institutions

Transparent, auditable safety decisions that make AI systems accountable.

Stakeholders



User Personas, Stories & Requirements

Kelly, 26- Data Engineer

Story: I want consistent dataset labels so that I can train and evaluate models reliably.
Challenge: BeaverTails and WildGuardMix use different labels.
Need: Map labels from both datasets to a common target taxonomy.

James, 35 - Trust & Safety Manager

Story: I want clear safety decisions so that I can understand why content is flagged.
Challenge: Identifying the type of harm in unsafe content.
Need: Provide safe/unsafe labels and relevant harm categories.

Evie, 34 - AI Product Manager

Story: I want to evaluate safety models so that I can balance safety and usability.
Challenge: Detecting harmful content without unnecessarily blocking safe content.
Need: Evaluate accuracy, false positives, and over-refusal.

Allen, 30 - End User (Affected Stakeholder)

Story: I want my legitimate requests to be accepted so that I can receive helpful responses.
Challenge: Safe requests may be incorrectly blocked.
Need: Avoid unnecessary blocking of legitimate requests.

End User Journey & Use Cases

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01

Submit Request

The user enters a prompt or request into the AI application.

02

Safety Check

The safety classifier analyzes the request according to the declared safety policy.

03

Inspect Results

The user receives a response or a refusal based on the AI application's safety handling.

04

Proceed / Revise

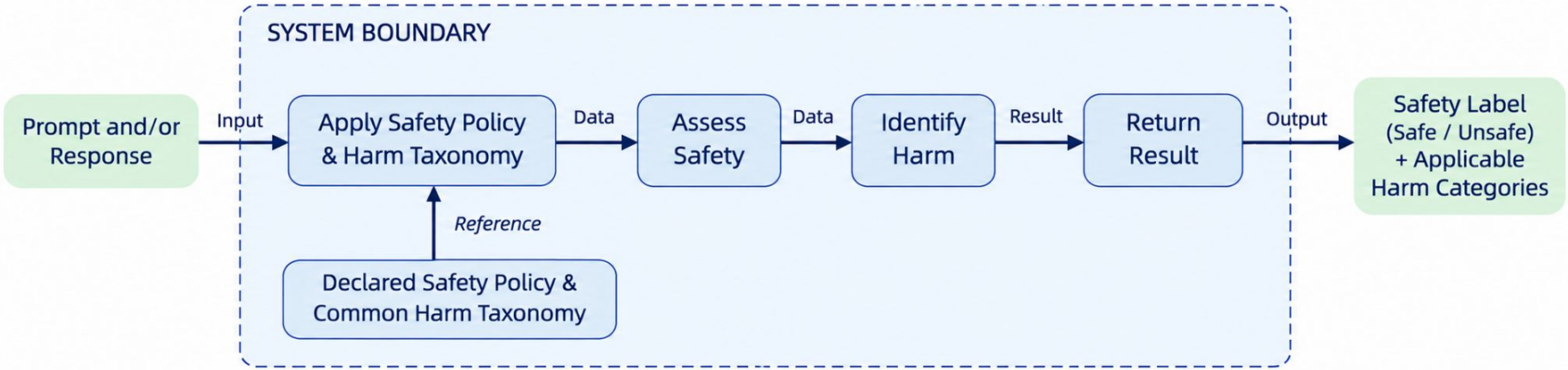
The user can continue the conversation or rephrase the request to receive a safe response.

END USER - USE CASES

UC-01 — Assess Prompt / Response

Classify the user's prompt or the AI-generated response and return a safety decision (safe/unsafe) with the applicable harm categories.

Functional Diagram



Proposed Functional View

Functional & Non-Functional Requirements

FUNCTIONAL

- **Assess Safety**
  Classify prompt/response as safe or unsafe.
- **Identify Harm**
  Return relevant harm-category codes.
- **Align References**
  Map labels to target policy.
- **Editable Taxonomy**
  Specify the 14-category taxonomy in the prompt.

NON-FUNCTIONAL

- **Reliability**
  Limit harmful misses, false positives and over-refusal.
- **Transparency**
  Make decisions inspectable, logged and reviewable.
- **Regulatory Compliance**
  Assess relevant risk and transparency obligations.
- **Reproducibility & Feasibility**
  Record experiment settings; track runtime and memory.

Risks

Technical

- **Over-refusal & False Positives**: Aggressive safety classification blocks legitimate prompts, reducing system utility.
  - **Mitigation:** Strictly minimize false positives by evaluating the fine-tuned model zero-shot and benchmarking it against the DeBERTa encoder baseline.
- **Decision Opacity:** Binary block/allow decisions lack explanatory reasoning, creating a non-debuggable black box.
  - **Mitigation:** Force the Llama 3.1-8B decoder to output the specific violated category code to ensure all decisions can be inspected and logged.
- **Resource Exhaustion:** Using an 8B parameter model (Llama 3.1-8B) as a classifier causes latency and high memory usage.
  - **Mitigation:** Utilize Parameter-Efficient Fine-Tuning (PEFT) (FR-06) and systematically measure memory and processing time under stated hardware conditions.

Risks

Governance and Security

- **Regulatory Non-Compliance:** The final guardrail fails to meet the emerging risk and transparency duties mandated by legislation like the AI Act.
  - **Mitigation:** Ensure regulatory readiness by embedding an editable taxonomy that clearly delineates prohibited content.
- **Repository Exposure:** Sensitive configurations or proprietary organizational requests are accidentally exposed to the public.
  - **Mitigation:** Confirm repository visibility constraints with the LINKS organization prior to open delivery on GitHub.

Data and Pipeline

- **Taxonomy Misalignment:** Training data from BeaverTails and WildGuardMix use different labeling conventions for prompt harms, response harms, and refusals, leading to a confused model.
  - **Mitigation:** Explicitly map and align labels from these disparate datasets to the target 14-category safety policy before fine-tuning

Work Packages

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WP1
Project Management
Lead **Alper**

WP2
Design
Lead **Nava**

WP3
Data Foundation
Lead **Rim**

WP4
Model Development
Lead **Rim**

WP5
Evaluation
Lead **Alper**

WP6
Communication
Lead **Nava**

Gantt Chart

ID	Work package / Task	NOVEMBER				DECEMBER				JANUARY			
		W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
WP1	**Project Management**												
1.1	Define work packages and timeline		D1.1										
1.2	Coordination and progress tracking												
1.3	Meetings with LINKS												
WP2	**Design**												
2.1	Problem, value proposition and objectives												
2.2	Stakeholders and personas												
2.3	Requirements and functional diagram		D2.1										
WP3	**Data Foundation**												
3.1	Explore BeaverTails and WildGuardMix												
3.2	Align harm taxonomies across datasets												
3.3	Prepare training, validation and test splits				D3.1								
WP4	**Model Development**												
4.1	Zero shot evaluation of existing guard models												
4.2	DeBERTa baseline per category					D4.1							
4.3	LoRA fine tuning of Llama Guard 3												
4.4	Hyperparameter tuning and target policy adaptation							D4.2					
WP5	**Evaluation**												
5.1	Define benchmark metrics												
5.2	Benchmark all models												
5.3	Error and over refusal analysis							D5.1					
WP6	**Communication**												
6.1	Checkpoint presentations		D6.1			D6.2			D6.3				
6.2	Final report and GitHub repo										D6.4		
MS	**Checkpoints and final submission**		CP1 11/11			CP2 02/12			CP3 21/12			FINAL 18/01	

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Thank you

Questions?